Introduction

Behind every major motion picture is a vast and diverse team of contributors who invested heavily in the film's production. Considering the significant time and effort it takes for most films to be built, as well as the oversaturation and growing competition within the film industry itself, it becomes increasingly important to make meaningful, intentional investments toward the production. Multiple factors can contribute to the future profitability of a film, so rather than relying on instinct, studio executives can instead utilize a predictive model to gain a better understanding of movie success trends. Namely, a model that uses several movie factors to predict the movie's revenue can be used to inform decisions. These insights will enable executives to allocate their investments into more promising films, as well as provide more context on whether an increased budget is justified.

Data

The data table provided includes 103 observations of movies, as well as 8 factors that describe each movie. The factor names, as well as what they describe, are listed in *Figure 1*. The original data table was then further manipulated to provide better insights. A new column specifying the director's age (*director_age*) was calculated with *birth_year*, and another factor was created to group the directors by decade (*director_age_group)*. The month, day, and year were extracted from the *production_date* and used to specify another factor that indicated the production season of each movie (*production_season)*. *Production_date* was also translated into a numeric value with the addition of a column called *production_day_number*, which counted the number of days that elapsed between the production day and January 1st, 1970. In order to easily

Factor name	Description
movie_title	The name of the movie
production_date	The day, month, and year that the movie was produced
genres	An alphabetical list of up to three genres that described the movie
runtime_minutes	Length of the movie in minutes
director_name	Name of the director
birth_year	The year the director was born
movie_averageRating	A quality rating given to the movie by viewers on a 1–10 scale, with 10 being the best
budget	The movie production budget in (nominal) dollars
gross	The movie's worldwide gross revenue in (nominal) dollars

Figure 1: Data table factor names and descriptions

keep track of the genres that belonged to each movie, dummy variables were added to the table. Genres that occurred at least 3 times were given their own column specified by "*D_*" followed by the genre. The nonfrequent genres were condensed into a *D_Other* column. Lastly, in light of the large values listed in the *gross* and *budget* variables, the natural log of both variables was taken and stored in columns titled *log_gross* and *log_budget,* which were used in place of *gross* and *budget* during exploratory data analysis and model selection.

Exploratory Data Analysis

First, the summary() function was used to report the minimum, maximum, and mean of each continuous variable to check for extreme or illogical values. Then, exploratory data analysis was conducted to investigate both the univariate and bivariate distributions of the variables. The director names, production seasons, and genres were plotted in a bar graph against their frequencies.

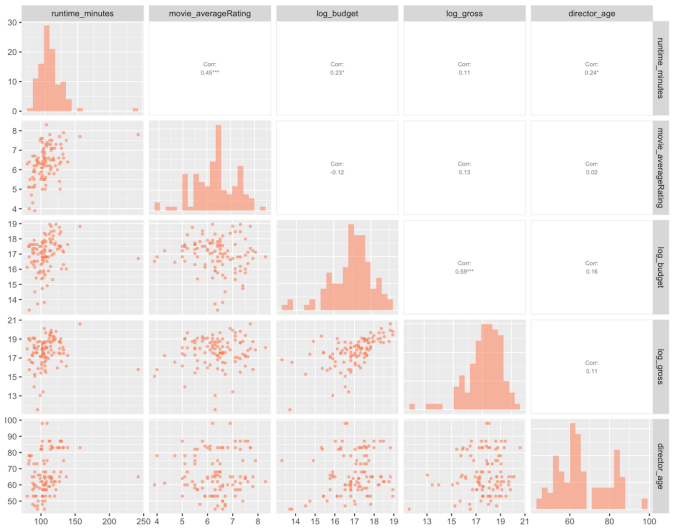

Figure 2: Scatterplot matrix bivariate and univariate distributions

Additionally, the bivariate and univariate distributions of the continuous variables, as well as their correlation with each other, were recorded using a scatterplot matrix. The strongest linear relationship was found between *log_budget* and *log_gross.* Additionally, *log_budget* exhibited a distribution most reminiscent of a normal distribution.

Model Selection and Details

To select the best model for predicting log_gross, the regsubsets() function was utilized to conduct an exhaustive test on the model space. To avoid computational infeasibility, the test was split into two main parts: main effects model exploration, and interactions model exploration. First, an exhaustive test was conducted to return the best regressor choices for main effects models with sizes ranging from one term to eight terms. Each main effects model was specified to have the comedy genre as a baseline. Among them, the best model was selected using the Bayesian Information Criterion, which is a conservative model selection criterion that favors simpler models. The best main effects model identified had three regressors: *movie_averageRating*, *log_budget*, and *D_Drama*.

Narrowing down the regressors to the ones chosen in the main effects model, another exhaustive test was conducted to investigate every possible combination of main effects and two-way interaction models. The Bayesian Information Criterion was also used to identify the three best models. The best model had regressors *movie_averageRating*, *log_budget*, and *D_Drama*, and a single interaction term *movie_averageRating:log_budget*. The second-best model was the main effects model. The third-best model had regressors *movie_averageRating*, *log_budget*, and *D_Drama*, and two interaction terms: *movie_averageRating:log_budget* + *log_budget:D_Drama.* Diagnostics were then run on each model. The two models that best satisfied the model's assumptions were then compared using an F-test, which found the second model to be more fitting in predicting *log_gross*.

Model Fit and Diagnostics

The following equation reports the best model selected to predict *log_gross*, where *MaR* stands for *movie_averageRating:*

$$log_gross_i = -42.5426 + 7.0902MaR_i + 3.3865log_budget_i - 0.8252D_{DRAMAi} - 0.3898(MaR_i * log_budget_i) + e_i,$$
$$e_i \sim N(0, \sigma^2)$$

In order to verify the model's validity, certain assumptions must be met. A linear relationship must exist between the predictors and the response, and the errors must be independent, normally distributed, and constant. Assessing the independence of errors is a multi-faceted process that includes determining how the data was collected. The remaining three assumptions can be tested by running diagnostics. In *Figure 3*, the "Residuals vs Fitted" diagnostics plot exhibits points that are randomly scattered around the zero horizontal line. This suggests a linear relationship between the predictors and the response variable, thus satisfying the linearity assumption. In the "Q-Q Residuals" plot, most points except for 3 anomalies fall roughly along the dashed 45-degree line. This suggests an approximately normal distribution of the error terms and adequately satisfies the normality assumption. The standardized residual values in the "Scale-Location" plot appear to be moderately scattered around the zero horizontal line. Although the variance appears to slightly increase for higher fitted values, the discrepancy is not a profound violation of the constant variance assumption.

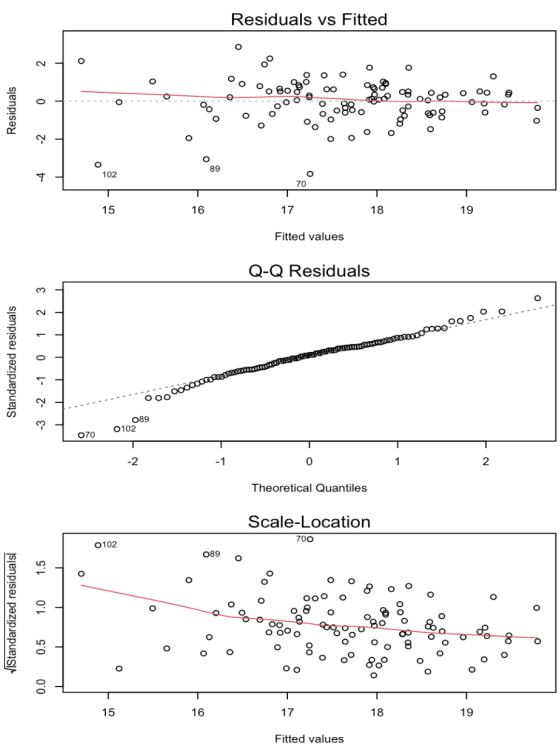

Figure 3: Linear model diagnostic plots

Interpretation

Figure 4 provides an interpretation of the regression coefficients. According to the table, the regression coefficient for budget represents the percent increase in gross revenue for each 1% increase in budget. Thus, a coefficient greater than one would represent revenue that grows faster than spending. A hypothesis test was conducted to determine whether the true parameter for *log_budget* was greater than one. The test returned a p-value of 0.01498408, which meant that for significance levels greater than 0.01498408, there was enough statistical evidence to claim that the coefficient for *log_budget* was greater than one.

Interpretation of Regression Coefficients

Response Variable: ln(gross)

Coefficient	Estimate	ln(gross) interpretation	Gross revenue interpretation
B_0 (Intercept)	−42.54	Expected ln(gross) for a non-Drama film with rating = 0 and budget = $1.	No meaningful real-world interpretation; serves as a baseline from which other effects adjust revenue.
B_1: Movie Average Rating	7.09	For non-Drama films, a 1-point increase in rating increases ln(gross) by 7.09, before accounting for budget interaction.	At a given budget, higher ratings substantially increase revenue, but the size of the effect depends on the budget.
B_2: ln(budget)	3.39	For non-Drama films at rating = 0, a 1% increase in budget raises ln(gross) by 0.0339.	A 1% increase in budget leads to about a 3.4% increase in gross revenue, holding movie rating constant.
B_3: Drama	−0.83	Drama films have ln(gross) 0.83 lower than non-Drama films, holding rating and budget constant.	Drama films earn about 56% less gross revenue than non-Drama films on average.
B_4: Movie Average Rating × ln(budget)	−0.39	As budget increases, the positive effect of rating on ln(gross) decreases.	Big-budget films gain less additional revenue from higher ratings than smaller-budget films.

Figure 4: Regression Coefficients and Interpretations

Although a promising correlation between budget and gross revenue was found, causal claims cannot be made to justify an increased budget unless the bias of the data is accounted for. Bias comes from the way the budget is not randomly assigned among the movies. Some movies are associated with famous actors or producers, boosting confidence in the movie's success and impacting spending. One way to reduce the bias would be to augment the table with more factors that could impact the budget. Another way to reduce the bias would be to apply the question to a smaller subset of the data. For example, one could want to investigate whether an increased budget is justified among movies with a certain similarity, like popularity or genre.

Prediction

Given a new film characterized by specific values of director name, production date, movie run time, movie rating, genre, and budget, the best model reveals the insignificance of using director name, production date, and movie run time for the prediction. Thus, the revenue can be predicted using only movie rating, genre, and budget. The predict() function was used to return a 95% confidence interval predicting the revenue of a new comedy film produced on June 15, 2013, with a run time of 2 hours, a rating of 7.0, and a budget of 10 million dollars. The fit returned to be $48,263,942, with a lower confidence limit of $5,044,977 and an upper confidence limit of $461,728,150. In other words, with 95% confidence, the true expected revenue of the film with the mentioned characteristics falls within the mentioned confidence limit.